TETHER: Tethered Causal Temporal Refinement for Frozen Surgical Stereo

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***Abstract*— Dense stereo is attractive for robotic surgery because it recovers metric geometry from the imaging the platform already carries, yet modern estimators are applied independently to each frame and can fluctuate under specularities, weak texture, tissue deformation, tool occlusions and camera motion. We ask whether temporal information can improve frozen stereo predictions without modifying the estimator, accessing its internals, or using future frames.**

We present TETHER, a causal plug-and-play temporal refiner for frozen stereo. At each frame, preceding disparity is motion-aligned to the current view using frozen SEA-RAFT, and a shared 38-channel external evidence representation of disparity, motion and image geometry feeds a 154,874-parameter reset-and-fusion head that predicts a bounded interpolation between the current stereo estimate and that aligned memory. No learned appearance features and no frozen backbone enter that evidence. Corrected recurrence is explicitly bounded and periodically re-anchored to independently predicted raw geometry.

A controlled development-set closure shows the learned policy is not explained by temporal smoothing: on SCARED-C D2, $H = 4$ reduces mean EPE by 3.69% against 0.59% for the best matched fixed/EMA blend and 0.87% for the best tuned unlearned rule on the raw-versus-memory residual, while flow-confidence weighting and ungated memory propagation degrade accuracy. The raw-versus-memory oracle exposes a 12.30% ceiling on D2, of which TETHER recovers 30.0%; continuous corrected-state recurrence increases EPE by 19.17%, supporting bounded memory.

One frozen $H = 4$ checkpoint improves all five evaluated stereo estimators on held-out SCARED-C D7, including CREStereo and Fast-FoundationStereo, excluded from temporal-head training. Mean backbone EPE decreases from 0.4921 to 0.4645 over three independently trained seeds ($5.61 \pm 0.42\%$), and Bad1 and RMSE decrease for every backbone. Zero-shot evaluation on five DRENDS recordings with all five estimators, two unseen in training, improves disparity EPE, Bad3 and metric depth in all 25 recording–backbone cells (-5.40, -19.51, -4.02, -3.47%), completing the transfer matrix across backbone and domain. Confining the fused output to the support the module already declares is load-bearing: without it, zero-padded out-of-bounds memory turns rare pixels invalid and produces a metric-depth tail previously attributed to the external domain — measured on the 142-channel configuration, which shares the warp and the output mask.

We additionally introduce a refinement-centric evaluation framework measuring geometry, temporal fidelity, introduced and recovered error, frame tails and selective risk under fixed support.

I. Introduction

Metric three-dimensional perception underpins camera navigation, tool–tissue distance estimation, augmented reality and deformation analysis in robot-assisted surgery, and stereo endoscopy is attractive because depth is recovered from images the platform already provides: the reference geometry used to validate it in this paper comes from structured light, CT and time-of-flight, none of which the deployed system needs at inference.

It is also hard. Endoscopic scenes carry weak texture, non-Lambertian tissue, specularities, blood, smoke, deformable anatomy, rapid viewpoint change and tool occlusion [18], [19]. Modern estimators [11]–[15] give increasingly strong frame-wise disparity, but nothing in a frame-wise model requires its predictions to cohere over time. Video stereo methods address this by building temporal reasoning into the model [1]–[4], at the price of assuming a temporal architecture, model-specific representations, future observations, or co-design with the estimator.

We study a different deployment setting: *can one causal temporal module be trained once and attached unchanged to heterogeneous frozen stereo estimators?* Backbone and temporal logic can then evolve independently, a new estimator replacing the previous one without the module needing its cost volume, hidden state, confidence head or training pipeline.

We keep one inductive bias from the video-stereo literature — decide whether motion-aligned history should influence the current estimate, and constrain the correction to a convex interpolation between the two — and redesign everything around it for a black-box interface: the integrated motion pathway becomes frozen SEA-RAFT, backbone-specific representations become a shared external evidence space, and the same head is evaluated unchanged across architectures.

The structure is deliberately constrained. The current estimate stays an anchor, temporal memory is geometrically aligned before use, the head controls how much to intervene but cannot leave the interval spanned by the two candidates, and corrected recurrence is periodically reset to raw geometry.

These restrictions raise a question: does the learned head add anything that simple temporal averaging would not? We answer it experimentally under a frozen development protocol. Matched fixed fusion, EMA, warped-memory propagation, and direct forward–backward confidence weighting recover only a small fraction of TETHER's gain. In contrast, a ground-truth oracle shows that the aligned memory contains substantially more useful information than the current policy exploits. This indicates that the core problem is not obtaining temporal evidence but deciding how much of it to use — and Sec. VI shows the head decides *how much* well and *where* no better than chance, which is what leaves the oracle gap open.

A second question is how such intervention should be evaluated. Smoothness is not correctness — a stale prediction can be smooth and wrong — and lower mean EPE does not reveal whether refinement introduces new bad pixels or rare catastrophic frames. We therefore evaluate refinement relative to the frozen upstream prediction and explicitly separate geometry, temporal fidelity, benefit, harm, and tail behavior.

Our contributions are three.

(i) We formulate surgical temporal stereo refinement as a *causal plug-and-play adaptation problem* over frozen, heterogeneous estimators, needing neither future frames nor backbone internals, and instantiate it as TETHER: a 154,874-parameter reset-and-fusion head over causal SEA-RAFT alignment, a shared 38-channel external evidence space of disparity, motion and image geometry, support-confined convex fusion and bounded corrected-state recurrence.

(ii) We measure one frozen checkpoint across the full backbone $\times$ domain matrix — five estimators, two of them excluded from temporal-head training, over development, held-out and out-of-domain splits — including the joint unseen-backbone, unseen-domain cell that is usually left unmeasured, and it improves every one of the fifteen.

(iii) We introduce a refinement-centric protocol under fixed, prediction-independent support that scores introduced error alongside recovered error, and show it changes conclusions: under a matched state machine, every fixed or EMA blend that recovers a meaningful share of the gain buys it by *raising* the bad-pixel rate, a trade the learned gate does not make and mean error does not reveal. Applied to our own claims it is equally unkind — degenerate history-replay controls win the no-reference consistency metric, and the head's own weight inverts its relation to harm across splits.

II. Related Work

a) Deep stereo matching.: Stereo has moved from cost-volume regression to iterative recurrent estimation [11], [12] and broad-domain zero-shot models [13]–[15]. Our work is orthogonal: we neither modify the estimator nor assume an architecture.

b) Temporal stereo and depth.: CODD introduced the reset/fusion mechanism we build on [1]; DynamicStereo [2], PPMStereo [4] and TC-Stereo [6] follow, all co-designed with the estimator, so a new backbone means a new temporal model. Match Stereo Videos exposes BiDAStabilizer as a plug-in on a frozen estimator [3], but propagates bidirectionally. A parallel line enforces consistency on monocular video depth by diffusion [24] or stabiliser networks [16]: a different output space, none causal at clip level. A further line adapts the stereo weights online [25], the opposite of our constraint.

c) What can actually be compared.: We are aware of no published method that is simultaneously causal, black-box and estimator-agnostic. CODD [1], whose two-gate decision we build on, is causal but not black-box: its fusion network consumes matching features from its own stereo network, which a cached third-party backbone does not expose. The 142-channel configuration of Sec. V-I began as the closest reproduction of it our interface admits — its projection replaced by our warp, and a frozen ResNet-18 standing in for the features it would have taken from the estimator — so the comparison against it in Table V is the nearest thing to running CODD here. The closest black-box method, StereoDiffusion [7], ships no implementation; neural disparity refinement [8] and blind video temporal consistency [9] publish code whose weights are no longer retrievable. Two methods are runnable here and we report both. BiDAStabilizer [3] refines a frozen estimator as we do, but bidirectionally, so it sees the future while we do not. TC-Stereo [6] is an integrated architecture rather than a refiner, and propagates its temporal state by a rigid reprojection from per-frame pose, intrinsics and baseline — privileged inputs our interface does not assume, but which SCARED-C does supply. We therefore run it as its authors intended (Sec. V-G), which measures what an estimator co-designed with its temporal path buys on this data, and what its rigid-scene assumption costs on deforming tissue.

d) Evaluating temporal refinement.: Stereo benchmarks report frame-wise geometry and video methods add consistency measures, but neither characterises an intervention: a prediction can grow smoother without growing more correct, and a lower mean error can hide newly created failures. Recovered and introduced error must therefore both be measured, which connects to selective prediction [17] and to the uncertainty literature asking when a prediction should be trusted [26]. Our contribution is not a new metric but a common protocol under fixed, prediction-independent support.

e) Surgical stereo geometry.: SCARED [18] and DRENDS [22] supply the endoscopic stereo and time-of-flight geometry we evaluate on; Sec. IV states what each can and cannot support. D4D [21] enters only as the substrate for a no-reference control, and SERV-CT [19] not at all: its pairs are non-consecutive, so temporal refinement on it is an exact identity and we report no result from it. Downstream reconstruction and tracking pipelines consume exactly the per-frame geometry we refine [20], [27], which is why we treat refinement as an intervention on that geometry.

III. Method

A. External Causal Interface

A frozen estimator S produces $d_t^{\mathrm{raw}}, M_t^{\mathrm{raw}} = S(I_t^L, I_t^R)$, positive-left disparity and its validity mask. The temporal module never reads the internal representation of S; it consumes only externally observable quantities — RGB, disparity, validity, optical flow and evidence derived from them. At time t the cache holds the preceding left image, the preceding raw disparity and validity, the previous fused state and its age; nothing from $t+1$ onwards exists. The map to learn is therefore $d_t^{\mathrm{fused}} = T_\theta(d_t^{\mathrm{raw}}, m_{t-1}, I_t^L, I_t^R, I_{t-1}^L)$.

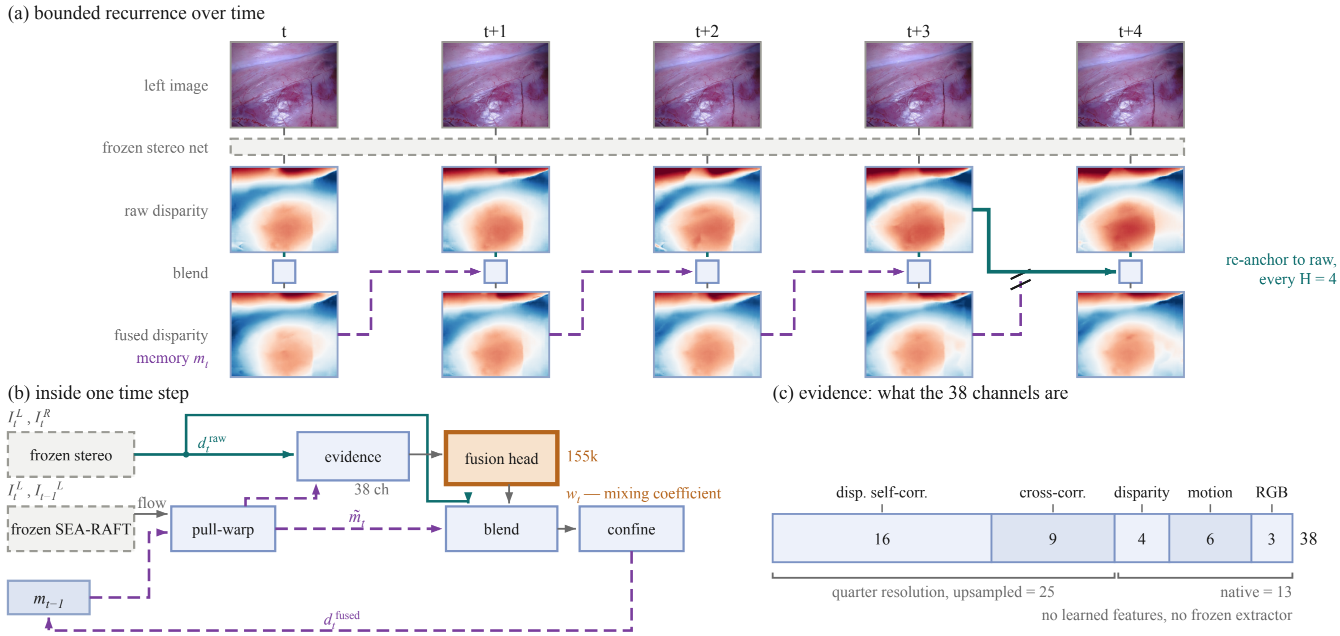

Fig. 1. TETHER at three scales. **(a)** Bounded recurrence: memory carries the *fused* disparity forward, and every $H{=}4$ frames the chain is cut and re-tied to raw geometry the frozen network produced independently (teal). The five frames are consecutive and the two disparity rows share one colour scale, differing by 0.126 px on average. One fitted plane is removed from all ten disparity panels: the surface is smooth at this resolution, a 7 px depth ramp carrying about 1 px of anatomy, so an absolute rendering shows a gradient and no scene. The same plane is subtracted from both rows, so the difference between them is untouched. **(b)** One time step; dashed grey is frozen, the single orange block is the only trained component and emits one scalar per pixel, w_t, rather than geometry. **(c)** The evidence space: none of the 38 channels carries ground truth, backbone identity or a future frame, and none is produced by a learned feature extractor.

B. Causal Motion Alignment

Frozen SEA-RAFT [5], [23] estimates both directions between the two already observed left images, $F_{t\to t-1} = F(I_t^L, I_{t-1}^L)$ and $F_{t-1\to t} = F(I_{t-1}^L, I_t^L)$; the second is used only to measure reliability. Bidirectional *flow estimation* between two observed frames is not bidirectional temporal inference.

Previous geometry is pulled onto the current grid, $\widetilde{m}_t(x) = m_{t-1}(x + F_{t\to t-1}(x))$, by bilinear sampling with zero padding and `align_corners=True`. Warp support $\mathcal{S}_t$ records whether the source coordinate lies inside the preceding image, and historical validity is sampled on the same grid. Pulling the reverse flow the same way, $\widetilde{F}_{t-1\to t} = \mathcal{W}(F_{t-1\to t}, F_{t\to t-1})$, gives the forward–backward residual $e_t^{FB} = \|F_{t\to t-1} + \widetilde{F}_{t-1\to t}\|_2$ and the reliability cue

$$C_t^{FB} = \mathcal{S}_t \exp\!\left(-e_t^{FB}/\rho_t\right), \qquad \rho_t = 0.5 + 0.01\left(\|F_{t\to t-1}\|_2 + \|\widetilde{F}_{t-1\to t}\|_2\right). \tag{1}$$

The zero padding in this warp is not a detail: it is what makes $\widetilde{m}_t$ exactly zero outside the source image, and Section V-J shows what happens when that value is allowed into the output.

C. Shared External Evidence

All stereo estimators are mapped into one common evidence representation:

$$C_t = \Phi\left(d_t^{\mathrm{raw}}, \widetilde{m}_t, I_t^L, I_t^R, I_{t-1}^L, F_{t\to t-1}, C_t^{FB}, \mathcal{S}_t, \widetilde{M}_t\right). \tag{2}$$

The head consumes one 38-channel tensor on the cache grid (Fig. 1c), 25 of whose channels are formed at quarter resolution and upsampled. Those 25 are disparity geometry: 16 disparity self-correlations of the raw and aligned candidates over a dilated 3×3 neighbourhood with the centre removed, and 9 raw-to-aligned cross-correlations over the same neighbourhood with the centre kept. The remaining 13 are formed on the cache grid directly: raw disparity, aligned memory, their signed and absolute residual, 6 motion components and 3 RGB.

Nothing in this space is learned and no network evaluates it. An earlier configuration added 104 further quarter-resolution channels built from a frozen ResNet-18 [10] — appearance self- and cross-correlations and feature-matching cost curves — for 142 in total; Sec. V-I reports it as an ablation, because it costs an extra 157,504 frozen parameters and three forward passes per frame and does not survive a three-seed comparison.

Normalisation is fixed before training: disparities clipped to $[0, 64]$ and scaled by 64, signed residuals to $[-16, 16]$ scaled by 16, flow on $[-32, 32]$ and magnitude on $[0, 32]$, ImageNet-normalised RGB. No ground truth, backbone identity or future frame enters this space.

D. Reset-and-Fusion Decision Head

The two-gate decomposition of the temporal decision, and the constraint that the correction be convex in the two candidates, are CODD's [1]; the architecture, the motion pathway, the evidence space, the supervision and the training are not.

The cache-grid pathway maps the 38-channel input to 24 channels through a 3×3 convolution, GroupNorm, SiLU and a residual block; a coarse pathway expands to 48 channels at one-quarter resolution followed by three 48-channel residual blocks.

A full-resolution reset branch predicts $r_t = \sigma(z_t^r)$, while the coarse fusion branch predicts $f_t = \uparrow_4 [\sigma(z_t^f)]$. Running the context branch coarsely is a compute choice, not a requirement: it costs about $16\times$ less than the full-resolution alternative, which Sec. V-I reports as an ablation rather than an adopted design.

The effective temporal weight is $w_t = r_t f_t$.

Temporal evidence is only defined where the aligned memory is usable, so the per-pixel *support* is the conjunction of current raw validity, aligned temporal validity and warp support,

$$S_t = M_t^{\text{raw}} \wedge \widetilde{M}_t \wedge \mathcal{S}_t, \tag{3}$$

— the warp support $\mathcal{S}_t$ of the previous paragraph, the raw validity and the sampled historical validity — and confine the intervention to it. Final disparity is

$$\boxed{d_t^{\text{fused}} = \begin{cases} (1-w_t)d_t^{\text{raw}} + w_t\widetilde{m}_t, & S_t = 1, \\ d_t^{\text{raw}}, & S_t = 0. \end{cases}} \tag{4}$$

Because $w_t \in [0,1]$, on the support

$$\min(d_t^{\text{raw}}, \widetilde{m}_t) \le d_t^{\text{fused}} \le \max(d_t^{\text{raw}}, \widetilde{m}_t), \tag{5}$$

and off the support the module is the exact identity on raw geometry: the refiner cannot synthesize disparity outside the interval spanned by its two candidates.

Applying (3) to the output, not only computing it, is load-bearing. The pull-warp uses zero padding, so an out-of-bounds sample returns $\widetilde{m}_t = 0$ *exactly*; an unmasked blend there drives d_t^{fused} to zero, producing an invalid disparity where the frozen estimator produced a valid one, and bounded recurrence carries it to the next re-anchor. Sec. V-J quantifies it on the 142-channel configuration, which shares this warp and this mask: invisible on SCARED-C, and the whole external-domain metric-depth tail.

The exact trainable parameter count is 154,874, with no frozen feature extractor in the loop; masking changes no parameter, no threshold and no training procedure.

Reset output bias is initialized to -2, while fusion is initialized with zero bias, producing conservative initial temporal intervention.

E. Bounded Corrected-State Recurrence

After refinement, $m_t = d_t^{\text{fused}}$.

This creates the feedback path

$$d_{t-1}^{\text{fused}} \to \widetilde{m}_t \to d_t^{\text{fused}} \to \widetilde{m}_{t+1}. \tag{6}$$

Continuous propagation can therefore transform previous alignment and fusion errors into future evidence.

We explicitly bound corrected-state lifetime. On a sequence boundary the first output is raw. After a re-anchor, previous raw disparity is used as state. Otherwise, the previous fused output is propagated.

The shipped policy permits four corrected-state uses before a raw re-anchor. Thus H denotes state lifetime rather than a temporal input window.

The method remains one-step causal: $\{I_{t-1}, I_t, m_{t-1}\} \to d_t^{\text{fused}}$.

F. Training Objective

For target disparity g_t let $e_t^{\text{raw}} = |d_t^{\text{raw}} - g_t|$, $e_t^{\text{mem}} = |\widetilde{m}_t - g_t|$ and $\Delta_t = e_t^{\text{mem}} - e_t^{\text{raw}}$, so $\Delta_t < 0$ marks pixels where aligned memory is the better candidate. Training support M is the intersection of GT coverage, current validity, aligned temporal validity and warp support.

The objective is $\mathcal{L} = \mathcal{L}_{disp} + \mathcal{L}_{reset} + \mathcal{L}_{fusion}$, unweighted, with $\mathcal{L}_{disp} = \mu_M[\text{Huber}_{\delta=1}(d_t^{\text{fused}} - g_t)]$, writing $\mu_A[\cdot]$ for the mean over pixel set A. The decision terms supervise the branches against Δ_t at native margins $\tau_r^n = 5\,\text{px}$ and $\tau_f^n = 1\,\text{px}$, scaled to the cache grid by 0.140625: writing $R^\pm$, $F^\pm$ for the sets where Δ_t exceeds $\pm\tau_r^c$, $\pm\tau_f^c$ and F^0 for $|\Delta_t| \le \tau_f^c$,

$$\mathcal{L}_{reset} = \mu_{R^+}[r_t] + \mu_{R^-}[1 - r_t], \tag{7}$$

$$\mathcal{L}_{fusion} = \mu_{F^+}[f_t] + \mu_{F^-}[1 - f_t] + 0.2\,\mu_{F^0}[|f_t - 0.5|]\,. \tag{8}$$

No explicit temporal-consistency objective is used: temporal behaviour is a consequence of the decision, never a training target.

IV. Experimental Protocol

A. Training

The temporal refiner is trained on temporally curated SCARED-C data derived from SCARED [18].

Training uses frozen predictions from S²M²-S [14], RAFT-Stereo [11] and Stereo Anywhere [13]; CREStereo [12] and Fast-FoundationStereo [15] are excluded from temporal-head training, and no backbone identity is ever given to TETHER.

Training uses D1, D3 and D6 with D2 for development and D7 held out, over 8,616 frames giving 8,606 causal pairs from three seen backbones, using non-overlapping $H = 4$ clips, batch size 4, 12 epochs of AdamW at 2×10^{-4} with weight decay 10^{-4} and gradient clipping 1.0. The 154,874 trainable parameters are selected on development fused EPE. That budget deserves scrutiny, since selection fires at epoch 8, 11 and 11 across the three seeds and two of those are the penultimate epoch. What the curves show is a plateau rather than a climb: from epoch 5 the mean epoch-to-epoch movement in validation EPE (0.0025–0.0046 across the three seeds) is comparable to everything still gained after it (0.0041–0.0054), and all three seeds end the run above their own minimum. The two are not strictly comparable — one is a single step, the other a cumulative gain over several — so this supports the weaker claim that the epoch of the minimum is uninformative, not that the curve has converged. **[TODO: convergence probe: the shipped model retrained for 100 epochs against the locked 12,**

from training_runs/convergence_probe_A2_no_learned_evidence_100ep. Eight times the budget, because doubling it would only move the question. Not eligible for selection; if it keeps improving that is a limitation to report, not a model to adopt.] One ablation (A4) diverges after epoch 10, which best-validation selection catches.

B. External Evaluation

a) DRENDS.: DRENDS contains dynamic robotic-endoscopy sequences with stereo imagery, calibration, and metric geometry [22]. We report five curated recordings — `Vid10_Liver_Med`, `Vid11_Liver_High`, `Vid12_Pancreas_Ext`, `Vid13_Pancreas_Med` and `Vid14_Pancreas_High` — 7476 frames, evaluated with all five stereo estimators. The zero-shot claim applies only to the SCARED-trained checkpoint: no DRENDS data enters training, checkpoint selection or threshold calibration.

Disparity reference is derived from metric ToF geometry through stereo calibration. It is therefore useful metric evidence but not independent stereo ground truth, and the ToF reference itself is temporally filtered.

C. Unified Temporal-Refinement Evaluation

A central requirement is that evaluation support cannot depend on the prediction: $M_{\text{eval}} = M_{\text{GT}} \cap M_{\text{protocol}}$.

Invalid predictions on valid support are penalized rather than silently removed.

The framework evaluates five dimensions.

a) Spatial geometry.: For disparity: EPE/MAE, median, RMSE, P90/P95/P99, maximum, invalid rate and Bad-$1/3/5$; for metric depth also Bad-$2/5/10$ mm, AbsRel, SqRel and $\delta_{1,2,3}$.

b) Temporal fidelity.: At horizons $k \in \{1, 2, 4, 8\}$ the framework defines grid-based temporal change error (DTCE), $|\Delta\text{prediction} - \Delta\text{target}|$ on fixed support, and, given correspondence, motion-compensated $\text{TEPE}_{\text{corr}}$ and prediction consistency, estimated-flow correspondences marked as proxies. DTCE scores agreement with the *reference* change, not stillness: a frozen prediction has $\Delta{=}0$ and is penalised by the whole of the reference motion, so the degenerate replay that wins the no-reference score of Sec. VII does not win this one. Every DTCE figure here is at lag $k{=}4$ unless stated, and we report the other lags where they disagree. We do not build on the motion-compensated family: it improves on all five D7 backbones at all four horizons but the two measures disagree on D2.

c) Intervention harm and benefit.: For $\delta e = e^{\text{fused}} - e^{\text{raw}}$, we measure

$$H^+ = \mathbb{E}[\max(\delta e, 0)], \tag{9}$$

$$B^+ = \mathbb{E}[\max(-\delta e, 0)], \tag{10}$$

$$\text{HUR} = P(\delta e > 0), \tag{11}$$

$$\text{BUR} = P(\delta e < 0). \tag{12}$$

For each bad-pixel threshold τ we additionally report NewBad, RecoveredBad, bad-rate delta, and identity preservation — the fraction of pixels already within τ that are still within it after refinement. It is a retention rate, not a count of untouched pixels: the blend moves almost every pixel on the support, and the question is whether it moves the good ones out.

d) Frame-level tails.: For each frame,

$$D_t = \frac{1}{|M_t|} \sum_{x \in M_t} \left(e_t^{\text{fused}}(x) - e_t^{\text{raw}}(x)\right). \tag{13}$$

We retain mean, median, P95, P99, worst degradation, and the fraction of degraded frames.

Where a risk score exists the framework also supports risk–coverage analysis and its calibration statistics [17]. Primary aggregation is equal-sequence macro averaging; pixel-weighted micro statistics are retained secondarily.

V. Results

A. Does the Learned Temporal Policy Matter?

We isolate the learned decision policy on D2. All variants share the frozen stereo predictions, the alignment, the support contract and the evaluation code; only the policy changes, and all run unconfined so the comparison shares one implementation (confinement, Sec. V-J, moves D2 EPE by $1.2{\times}10^{-5}$ px). Figures here pool all five backbones and every D2 sequence before taking a ratio, so 0.276202 is one raw mean over the fifteen backbone–sequence cells; reductions per backbone or per arena average per-cell relative reductions instead.

Raw D2 EPE is 0.276202. Table II gives the closure; the fifteen policies are the horizon grid $H \in \{1, 2, 4, 6, 8\}$ and continuous recurrence for the learned head, fixed $w \in \{0.1, 0.2, 0.3, 0.5\}$ rising monotonically to 0.59%, EMA over two and three frames, forward–backward confidence as the weight, aligned-memory propagation, and warped raw previous at $H{=}1$. A tuned two-parameter rule on the raw-versus-memory residual, swept over seventeen (α, τ) points, reaches 0.87% at an interior setting. Per backbone, against whichever fixed or EMA policy is best *for that backbone*, the ratio ranges $3.9\times$ to $9.9\times$ and is $5.0\times$ excluding Stereo Anywhere: the absolute gain depends on the estimator, the margin over smoothing does not.

Two rows carry more than the mean. Forward–backward confidence is the informative failure — one of TETHER's own cues, yet used directly as the weight it worsens EPE by 2.17% and Bad1 by 32.35%. And the matched smoothing policies buy mean error with bad pixels, the more so the harder they blend: the two weakest fixed settings move neither metric by a hundredth of a point, $w{=}0.3$ costs 0.18% more bad pixels for 0.49% of EPE, the best fixed blend costs 2.64% for 0.59%, and EMA3 costs 8.88% for 0.30%. Averaging trades tail correctness for mean error as soon as it does real work; the learned gate does not make that trade, and mean EPE alone does not show it.

B. Why $H = 4$, and Why Recurrence Must Be Bounded

Recurrence must be bounded: with no re-anchor the intervention inverts — EPE degrades by 19.17%, Bad1 by 143.48% and temporal stability by 28.13% (Table II). Within

TABLE I

ONE FROZEN TETHER CHECKPOINT, SEED 0, ACROSS FIVE ESTIMATORS AND THREE ARENAS; EVERY EPE AND BAD1 CELL IS RAW→TETHER, LOWER IS BETTER. TWO ESTIMATORS WERE EXCLUDED FROM TEMPORAL-HEAD TRAINING, SO THE LOWER-RIGHT BLOCK IS THE JOINT SHIFT USUALLY LEFT UNMEASURED. DTCE IS THE REDUCTION IN GRID-BASED TEMPORAL CHANGE ERROR AT LAG k=4, IN %, HIGHER IS BETTER, AVERAGING PER-SEQUENCE RELATIVE REDUCTIONS AS EVERY PER-BACKBONE REDUCTION IN THIS PAPER DOES; ON THE DEVELOPMENT SPLIT IT IS BACKBONE-DEPENDENT AND WE REPORT IT ONLY WHERE IT IS CONFIRMATORY (SEC. V-B). SEC. V-E GIVES THE THREE-SEED SPREAD.

		D2 development		D7 held out			DRENDS, out of domain		
Backbone	Status	EPE	Bad1 (%)	EPE	Bad1 (%)	DTCE	EPE	Bad1 (%)	DTCE
S^2M^2-S	seen	.2704 → **.2658**	1.39 → **1.27**	.5176 → **.4992**	6.44 → **6.03**	6.5	1.3477 → **1.2812**	62.55 → **58.98**	11.3
RAFT-Stereo	seen	.2691 → **.2623**	1.60 → **1.43**	.4671 → **.4296**	6.09 → **5.75**	7.3	1.3125 → **1.2371**	58.83 → **55.19**	8.5
Stereo Anywhere	seen	.3022 → **.2779**	2.18 → **1.74**	.4774 → **.4408**	6.23 → **5.77**	7.9	1.3685 → **1.2834**	62.73 → **57.99**	13.9
CREStereo	unseen	.2668 → **.2583**	1.57 → **1.40**	.5165 → **.4912**	6.35 → **6.05**	6.4	1.3027 → **1.2296**	55.17 → **52.21**	11.8
Fast-FoundationStereo	unseen	.2725 → **.2658**	1.60 → **1.44**	.4819 → **.4645**	5.92 → **5.65**	4.3	1.2838 → **1.2228**	54.88 → **52.06**	9.6

TABLE II

MATCHED-POLICY CLOSURE ON D2, ALL FIVE BACKBONES AND ALL SEQUENCES POOLED BEFORE TAKING EACH RATIO; REDUCTIONS IN %, HIGHER IS BETTER, DTCE AT LAG k=4. EVERY ROW SHARES THE FROZEN PREDICTIONS, THE ALIGNMENT, THE SUPPORT CONTRACT, THE H=4 STATE MACHINE AND THE EVALUATION CODE — ONLY THE POLICY CHANGES, AND THE SIX ROWS ABOVE THE RULE LOAD NO CHECKPOINT. THE FULL GRID OF FIFTEEN POLICIES IS IN THE TEXT. MEAN EPE IS NOT THE WHOLE STORY: THE SMOOTHING POLICIES BUY IT WITH BAD PIXELS, AND THE HARDER THEY BLEND THE WORSE THAT TRADE BECOMES.

Policy	EPE	Bad1	$\text{DTCE}_{k=4}$	NewBad1
Fixed w=0.5 (≡ EMA2)	0.59	−2.64	6.26	0.214
EMA, three frames	0.30	−8.88	4.88	0.379
FB-confidence as the weight	−2.17	−32.35	−7.10	0.831
Warped raw previous, H=1	−0.03	−6.39	−4.33	0.373
Aligned-memory propagation	−3.56	−41.70	−13.55	1.072
Unbounded corrected recurrence	−19.17	−143.48	−28.13	2.858
TETHER, H=4	**3.69**	**12.84**	**4.99**	**0.063**
Raw-vs-memory oracle (ceiling)	12.30	20.91	9.70	0.000

the bounded range EPE alone does not select an operating point — it improves from 2.49% at H=1 through 3.69% at H=4 to 3.73% at H=6 before falling to 3.52% at H=8. EPE does reject H=1, by 1.24 points; among the three horizons it does not separate, the spread is 0.21 — while newly broken pixels rise monotonically with the horizon, from 0.052% to 0.128%. We froze H=4 before D7 opened, on that risk–gain trade rather than on EPE, and it is the last horizon at which grid-based temporal change error also improves at lags $k \in \{2, 4, 8\}$: 4.99% at H=4 against −3.64% at H=6 for k=4.

Two caveats belong with that, because neither is visible in the pooled number. At k=1 DTCE improves at every horizon we tried, so the sign change is a property of the longer lags and not of the metric. And on D2 the improvement is carried by one backbone and is *negative* for three: +14.0% for Stereo Anywhere and +0.8% for CREStereo against −0.3%, −2.0% and −2.3% for the rest. On both confirmation splits it is positive for every backbone (Table I), which is why we report it there and rest the horizon choice on breakage rather than on stability.

C. Cross-Backbone Development Behaviour

$H = 4$ improves all five estimators in mean D2 EPE (Table I), from 1.41% to 8.19% averaging per-sequence reductions — 1.70% to 8.04% from the pooled cells the table prints — and all six unseen-backbone sequence groups improve; on D7 the mean excluding Stereo Anywhere is 5.15%. One D2 cell worsens, S^2M^2-S on keyframe 2, by 0.28%.

D. Held-Out Cross-Backbone Confirmation

We next evaluate the already frozen $H = 4$ policy on held-out D7. All five backbones improve, including both unseen during temporal-head training: mean EPE decreases 0.4921 → 0.4651, or 5.49% pooling before the ratio and 5.40% averaging per-cell reductions.

The improvement is not restricted to the mean: Bad1 and RMSE decrease for every backbone on both splits, and P99 and maximum error in all ten backbone–split cases, most strongly for Stereo Anywhere on D2 (33.27 to 13.03 px). InvalidRate stays zero in all ten SCARED-C backbone–split evaluations, a property of SCARED-C not of the module: Sec. V-J shows an unmasked output does introduce invalid predictions out of domain, which the support contract of (4) prevents.

The EPE gain is therefore not bought by a worse bad-pixel rate or a broader tail. New bad pixels do appear and are measured below. The joint unseen-backbone / unseen-domain cell is the lower-right block of Table I, read in Section V-H.

E. Seed Variance, and What It Reveals About Selection

We pre-registered a seed study before any result existed: only the seed varies, sequences are the resampling unit, and selecting the best seed is prohibited.

Pooled over five backbones the relative Bad1 reduction is 12.73 ± 0.32% on D2 and 5.84 ± 0.22% on held-out D7; EPE gives $4.13 \pm 0.38\%$ and $5.61 \pm 0.42\%$, where ± is the standard deviation over the three seeds throughout this paper and *range* always means min–max. A paired bootstrap over the 15 and 20 backbone–sequence cells gives $[8.96, 16.60]$ and $[4.00, 8.71]$ for Bad1 and $[2.55, 5.67]$ and $[4.03, 7.28]$ for EPE, positive in every resample. We report those intervals as descriptive only: the cells share three and four underlying sequences, so they are not independent draws and the interval

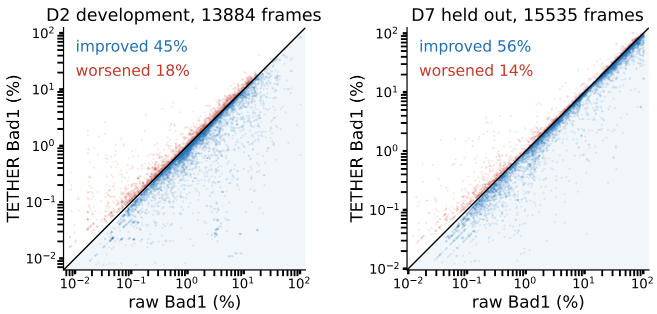

Fig. 2. Per-frame Bad1, raw against refined, for *every* scored frame of each split — five backbones, all sequences, no selection. Points below the diagonal are improved. The worsened band hugs the diagonal while the improved mass extends far from it: the frame-level count-versus-magnitude asymmetry behind B^+/H^+. On held-out D7, 56% of the 15,535 frames improve against 14% worsened; on development D2 the split is 45% against 18%, so the frame-level margin is wider on the split that took no part in any choice.

is narrower than the evidence warrants. The distribution-free statement the splits can carry is that the effect is positive in $3/3$ and $4/4$ sequences, a sign test at $p = 0.125$ and $p = 0.0625$.

The spread is the strongest argument for the shipped configuration, and it is not what we found first. The 142-channel ablation is stable on the held-out split (5.15 ± 0.34 Bad1) and unstable on development: its seeds fall monotonically, $10.78 \rightarrow 8.67 \rightarrow 5.75$, a 5.03 point spread, while the ordering inverts on D7 and the spread collapses to 0.69. We first read that as selection optimism, since checkpoints are selected on best D2 fused EPE and optimism should sit in the split used to select. Removing the learned evidence removes the instability instead: 0.32 points of D2 spread against 2.53, eight times tighter, with no change to the selection rule. The instability was the evidence space, not the protocol.

F. Refinement as an Intervention

Across both splits and all five backbones TETHER recovers 3.8–11.1$\times$ as many Bad1 pixels as it introduces, B^+/H^+ between 1.39 and 3.25, identity preservation above 99.85%: it repairs far more than it breaks (Fig. 2).

G. Head-to-Head Against a Published Stabiliser

BiDAStabilizer [3] is the closest published plug-in stabiliser we could run (Table III). We compare on DRENDS because only there is *neither* method in domain: we train on ex-vivo porcine tissue, it has never seen ex-vivo liver and pancreas manipulation against a time-of-flight reference. The shared input is its own RAFT-Stereo *robust* prediction, so TETHER is driven on that exact array over all five recordings (7476 frames) under one support neither method influences. Two asymmetries run in opposite directions and both belong here. The baseline is bidirectional and consumes future frames, which our setting forbids. We, on the other hand, are trained on surgical endoscopy and it is running weights fitted to natural-image stereo, so "out of domain" is not symmetric: neither has seen DRENDS, but only one of us has seen this kind of scene. The same holds of TC-Stereo below. We therefore read the margin as an upper bound on our advantage, and an in-domain baseline remains the comparison we cannot run.

TABLE III

OUT-OF-DOMAIN COMMON SUPPORT, FIVE DRENDS RECORDINGS, 117.4M PIXELS; *neither* METHOD HAS SEEN THIS DOMAIN. EVERY ROW REFINES THE SAME FROZEN RAFT-STEREO *robust* PREDICTIONS; LOWER IS BETTER. DTCE_k IS THE TEMPORAL CHANGE ERROR AT LAG k, IN PX, ON THE SAME SUPPORT. THE TWO BEST COLUMNS BELONG TO DIFFERENT ROWS, AND THAT IS THE FINDING: THE BIDIRECTIONAL BASELINE IS THE MORE TEMPORALLY STABLE METHOD AND THE LESS ACCURATE ONE.

Method	Causal	EPE	Bad1 (%)	Bad3 (%)	RMSE	DTCE_1	DTCE_4
Raw	—	1.199	44.92	4.54	1.876	0.242	0.340
BiDAStabilizer	no	1.175	44.81	4.32	1.776	**0.085**	**0.214**
TETHER	**yes**	**1.131**	**43.39**	**3.94**	**1.710**	0.139	0.294

The regime is hard and the Bad1 column says so: DRENDS ground-truth disparity averages 9.89 px and spans 8.42–11.89 between the 5th and 95th percentiles, so a raw EPE of 1.199 px is a third of the scene's disparity spread and a 1 px threshold rejects 44.9% of pixels. Not a property of one dataset: SCARED-C D7 keyframe 3 sits at raw EPE 1.18 px on comparable disparities. What follows compares two refiners of a weak prediction, not a claim that either recovers a good one.

Out of domain for both, TETHER removes between 1.7 and 14 times as much error as the bidirectional baseline, on all four metrics we score, while being the only causal entry and carrying 62$\times$ fewer trained parameters (155k against 9.64M). That ratio is about training cost, not about deployment: our system also runs frozen SEA-RAFT, 8.88M parameters and 26.2 of our 28.9 ms, so at system level the two are comparable in size and we make no efficiency claim. The largest of the four ratios, 14$\times$ on Bad1, divides our 1.53 points by the baseline's 0.11 and is not a stable statistic; the absolute deltas are the reportable form. It is not uniform, and the exception is worth naming: on Vid13 TETHER is 0.022 px behind the baseline and 0.013 px *worse than the raw prediction it refines*, the one recording of five where the intervention is a net harm and the bidirectional method is not. In domain the ordering is the same and the margin is far wider: on the D2 common support TETHER takes EPE $0.3029 \rightarrow 0.2889$ and Bad1 $2.09 \rightarrow 1.80\%$ where the baseline reaches 0.3021 and 1.99%, a 4.6% reduction against 0.3%. We report the neutral arena instead because that margin is measured where we are in domain and it is not.

The temporal measure reverses the ordering, and we report it for that reason. On the arena of Table III, change error falls 64.9% for the bidirectional baseline against 42.5% for us at lag 1 ($0.242 \rightarrow 0.085$ against 0.139), and 37.2% against 13.8% at lag 4; the ordering is the same in each of the five recordings and at every lag we measured. In domain it does not change either: on D2 the reductions are 52% against 15% at lag 1 and 24% against 4% at lag 4. A method that sees the future is markedly more temporally stable than one that does not, which is what should happen, and it is simultaneously the less accurate of the two on every geometric metric here. That is the paper's own warning turned on its own comparison: smoothness is not correctness, and a reader given only a consistency score would rank these two the wrong way round.

TC-Stereo [6] is the integrated-architecture reference, and what follows is not a claim that it is a weaker method. We run its released SceneFlow weights, its most general pretraining, unmodified and at its own iteration count, on surgical video it has never seen and was never adapted to. It is therefore doubly outside its operating conditions: out of its training domain, and — because its temporal path warps the previous disparity by a *rigid* 6-DoF reprojection from per-frame pose, intrinsics and baseline — outside its stated scene assumption as well. Its frame path already scores worse than the frozen prediction we refine, before any temporal component is involved, which is the size of the domain gap rather than of the architecture. SCARED-C supplies corrected per-frame poses — producing them is what the dataset is for — so the path is runnable here, and we run it. The pose frame is the one thing that must be right: SCARED-C poses live in each frame's native camera while the warped disparity is rectified, so the pose is conjugated by R_1. A self-check that warps ground-truth disparity between two frames with the method's own operator scores 0.0202 px under this convention against 3.64 and 2.07 px for the two alternatives, which is how we know it is the intended one. On the same D2 support, 18.0M pixels over all three keyframes, its frame path alone scores EPE 0.524 and Bad1 9.05%; enabling its temporal path makes both *worse*, 0.622 and 10.37% — against 0.303/2.09% raw and 0.289/1.80% for us.

The temporal measure is where the difference is categorical, and it is the reason we report it for the competitors and not only for ourselves. Grid-based temporal change error rises from 0.073 to 0.406 px at lag 1 and from 0.135 to 0.592 at lag 8: between four and six times *less* temporally consistent than the frozen prediction it refines, at every lag. A method built for temporal consistency, run as its authors specify with a pose convention validated to 0.02 px, degrades temporal consistency more than fourfold at every lag on this data. The rigid-scene assumption behind its warp is what deforming tissue violates, and a rigid warp of a deforming surface introduces exactly the frame-to-frame disagreement the metric counts. That is a property of the method meeting this data rather than a defect of the method: a co-designed temporal architecture encodes assumptions about the scene, and assumptions do not transfer even when weights are retrained. That is the argument for a refiner told nothing about scene geometry, not an argument that this particular method is poorly built. A TC-Stereo trained on deforming surgical tissue is an experiment nobody has run, ours included, and we make no claim about how it would do.

H. External Zero-Shot Geometry on DRENDS

We evaluate the frozen checkpoint on five DRENDS recordings with all five estimators, two excluded from temporal-head training, 7476 frames each. Nothing is adapted, retrained or tuned for this domain. Disparity EPE, Bad3, metric depth MAE and metric depth RMSE improve in *every* one of the 25 recording–backbone cells (Table IV), by -5.40%, -19.51%, -4.02% and -3.47% on average. The two unseen estimators are not distinguishable from the three seen ones, which is the claim.

TABLE IV

ZERO-SHOT DRENDS, ONE SCARED-TRAINED CHECKPOINT, FIVE RECORDINGS $\times$ FIVE ESTIMATORS $\times$ FOUR METRICS. RELATIVE CHANGE IN %, NEGATIVE IS BETTER; EACH ENTRY IS THE MEAN OVER THE FIVE RECORDINGS OF THE PER-RECORDING RELATIVE CHANGE. EVERY ONE OF THE 25 RECORDING–BACKBONE CELLS IMPROVES ON EACH OF THE FOUR METRICS. THE FOUR ARE NOT INDEPENDENT EVIDENCE — DEPTH IS THE SAME DISPARITY FIELD THROUGH ONE CALIBRATION — SO WE COUNT CELLS, NOT CELL-METRIC PRODUCTS. NOTHING IS ADAPTED, RETRAINED OR TUNED FOR THIS DOMAIN, AND TWO OF THE FIVE ESTIMATORS WERE NEVER SEEN IN TEMPORAL-HEAD TRAINING.

Backbone	Status	EPE	Bad3	Depth MAE	Depth RMSE
S^2M^2-S	seen	-4.86	-18.44	-3.65	-3.06
RAFT-Stereo	seen	-5.63	-20.71	-4.22	-3.58
Stereo Anywhere	seen	-6.21	-23.54	-4.60	-4.09
CREStereo	unseen	-5.62	-19.07	-4.12	-3.67
Fast-FoundationStereo	unseen	-4.70	-15.80	-3.50	-2.97
All 25 cells		-5.40	-19.51	-4.02	-3.47

Residual variance is set more by the recording than by the estimator, but not uniformly: across the five backbones the EPE change spans 0.87 points on Vid10 and 1.03 on Vid11, against 3.18 on Vid12 and 3.29 on Vid13. Where the estimator matters, it matters as much as the recording does.

Averaged over backbones, the mean EPE reduction is 3.96% for seen and 2.67% for unseen estimators on D2, 5.80% and 4.79% on held-out D7, and 5.57% and 5.16% on DRENDS: every combination is positive. We do not read an ordering into it. On D2 the seen advantage is carried entirely by Stereo Anywhere, and without it the seen mean falls to 1.85% against 2.67% unseen — the ordering reverses. Five backbones cannot support a claim about which axis costs more, and we withdraw the one we drew. This establishes zero-shot geometric transfer under a ToF-derived metric reference; it does not establish universal cross-domain robustness, and DRENDS remains a non-independent reference.

I. Ablating Our Own Design Decisions

Four design choices were ablated under a pre-registration naming held-out Bad1 as the endpoint and the three-seed spread (0.37 points) as the threshold below which a difference is not read; A4 remains a single run. A separate later registration governs *promotion* of a variant to the shipped model and decides that on development only, which is how the 38-channel head was promoted; the two rules are distinct and we apply each where it belongs.

The result (Table V) is not the one we expected, and it is why the shipped model is the smaller one. We began where

TABLE V

HELD-OUT RELATIVE REDUCTION (%), NOT ERROR: HIGHER IS BETTER. THE FOUR DEVIATIONS WERE PRE-REGISTERED AGAINST THE ORIGINAL 142-CHANNEL CONFIGURATION; DROPPING ITS LEARNED EVIDENCE IS WHAT TETHER NOW IS, SO THE SHIPPED MODEL IS THE TOP ROW AND THE CONFIGURATION IT WAS ABLATED FROM IS THE FIRST ROW BELOW; THE INDENTED ROWS ARE DEVIATIONS FROM THAT CONFIGURATION. THREE-SEED MEANS WHERE MARKED.

Configuration	Evidence	D7 EPE red.	D7 Bad1 red.
TETHER (3 seeds)	38 ch	$\mathbf{5.61 \pm 0.42}$	$\mathbf{5.84 \pm 0.22}$
+ learned evidence (3 seeds)	142 ch	5.06	5.15 ± 0.34
no appearance	78 ch	4.93	4.88
full-res context (3 seeds)	142 ch	5.24 ± 0.52	5.68 ± 0.16
unconstrained output	142 ch	0.05	6.02

the video-stereo literature does, with a learned feature extractor: a frozen ResNet-18 [10] supplying appearance self- and cross-correlations and feature-matching cost curves, 104 channels on top of the 38 geometric ones. That configuration is the one the four deviations were pre-registered against, and three of them sit above it on the held-out endpoint, so none of the decisions they undo is load-bearing. Removing the learned evidence entirely was then given the same three-seed treatment under a rule fixed before the extra seeds were trained — decided on D2 Bad1, with D7 reported afterwards and taking no part in the choice — and it stays ahead on both endpoints with a spread eight times tighter (0.32 against 2.53 points of D2 Bad1). Out of domain the same ordering holds: on the three backbones carrying three seeds of each configuration, DRENDS EPE falls by $5.20 \pm 0.13\%$ against $4.83 \pm 0.15\%$, the two seed ranges not overlapping.

The learned evidence is not what makes the module work; it is what makes its training unstable, and it costs 157,504 frozen parameters, three forward passes per frame and 13.8% of the runtime to do so. Its apparent advantage on the development closure (4.05% against 3.69%) is a seed-0 artefact: over three seeds TETHER leads there too, 4.13% against 3.55%. Two rows need the registered rule read out loud, because it does not favour us. Running the context branch at full resolution instead of a quarter reduces held-out Bad1 by $5.68 \pm 0.16\%$ against the base configuration's $5.15 \pm 0.34\%$ over three seeds each: it *wins* the registered endpoint, by 0.54 points against a 0.37-point threshold, an outcome the registration did not enumerate. We do not adopt it, and the ground is cost rather than accuracy — $16\times$ the context-branch compute — which the registration required us to report if this happened. Ablating convexity (A4) posts the largest held-out Bad1 reduction in the table, 6.02%, on one seed, and we reject it on EPE, which the registration names as reported but not as the endpoint; that reading was chosen after seeing the result and we mark it as such. Its registered consequence we do honour: the falsification clause said that if A4 matched or beat the canonical model on held out, the claim attributing cross-backbone transfer to the convex action space must be withdrawn rather than softened. It is withdrawn — that claim appears nowhere in this paper.

J. The Support Contract Is Load-Bearing

The results above use the support-confined output of (4); the ablation is a single line, with checkpoint, head, evidence, policy and reset schedule identical. It was measured on the 142-channel configuration, and concerns the output mask and the pull-warp padding rather than the evidence space. On DRENDS the difference is categorical. Unmasked, depth RMSE *regresses* in every cell evaluated, by $+110.4\%$; confined, it improves in every cell, by -3.2%. The two arms are not scored on the same set — the unmasked run covers 14 of the 15 backbone–recording cells, missing RAFT-Stereo on Vid14 — so the comparison is directional and the magnitudes are not paired. Disparity follows more quietly, EPE from -4.30% to -5.00%, and invalid refined pixels go from up to 1.3×10^{-5} to exactly zero.

The cause is that zero-padded out-of-bounds memory enters the blend as $\widetilde{m}_t = 0$, the fused disparity collapses there, and bounded recurrence carries it to the next re-anchor. The offenders are rare and bursty — 300 of 38.8M pixels on one recording, in runs of at most H frames — and the aligned memory of every one is exactly 0.0. Since an invalid prediction on valid support is penalised rather than discarded, those few hundred pixels are the entire source of the depth tail previously attributed to the external domain.

On SCARED-C the same ablation is nearly inert, again on the 142-channel configuration: D2 unchanged at 4.05%, D7 from 4.64% to 4.57%. The contract is therefore not free — it costs a little in-domain gain and buys well-posed behaviour out of domain — and it is almost invisible to anyone validating in domain only, which is what makes it worth stating.

VI. ANALYSIS

The closure isolates four requirements. Alignment is necessary but not sufficient: the aligned memory carries a 12.30% oracle ceiling, yet using it indiscriminately fails — warped raw memory yields no gain, and both FB-confidence weighting and full recurrent replacement worsen EPE. The mechanism is alignment plus learned graded intervention plus bounded recurrence, not smoothing. The head never solves stereo matching: it receives two meaningful candidates and chooses between them.

a) What each branch of $w_t = r_t f_t$ does.: Each factor can be suppressed at inference with the trained head unchanged, which separates what the reset gate contributes from what the fusion weight contributes. Fusion alone keeps most of the gain, 3.49% against 3.69%, but breaks a third more pixels (0.084% NewBad1 against 0.063%) and drops B^+/H^+ from 1.85 to 1.72. Reset alone keeps 2.23% and breaks 0.211%, more than three times as many. The two factors therefore do different jobs: the fusion weight sets how much is gained, the reset gate sets how little is broken, and a paper reporting only the product and only mean error would have seen neither. On the 142-channel configuration the same decomposition is flatter in EPE (4.01% against 4.05%) and the count asymmetry is the whole of it, so the reset gate earns more of the mean gain in the shipped head than in the one it replaced.

Ablating the convexity constraint (A4) collapses mean EPE gain to 0.05% while improving RMSE and worst case: the failure is broad degradation of pixels already close, not occasional large error, so the constraint buys mean accuracy rather than safety from large corrections.

Two limits are worth stating. Against the raw-versus-memory oracle of Table II, TETHER recovers 30.0% of the available gain on D2 as a ratio of the pooled quantities, 32.9% averaging per-cell recovery, so the open problem is deciding *where* to intervene. And the effective weight w_t, recovered from the convex form, is an intervention magnitude rather than an uncertainty: it ranks harmful intervention at AUROC 0.393–0.408 on D2 and 0.488–0.507 on held-out D7, below chance on the split it was tuned on and at chance on the split it was not. The 142-channel configuration is the same shape and further from chance (0.366–0.381 and 0.496–0.506), so this is a property of the formulation and not of the evidence space. It does not contradict the intervention ratios, which factorise into a count and a magnitude term needing no per-pixel ranking: on this analysis' own support B^+/H^+ is 2.03 on D2 and 2.86 on D7, and per-cell oracle recovery rises from 32.9% on D2 to 42.7% on held-out D7. A policy can therefore be good on average and uninformative per pixel, which is what makes the oracle gap a ranking problem rather than a capacity one.

VII. Discussion and Limitations

a) What the splits support.: The closure is development work on D2: it supports mechanism analysis and the pre-frozen $H = 4$ operating point, not confirmation. Transfer is shown along both axes and their combination, but one external domain does not establish universal robustness, and DRENDS supplies a temporally filtered ToF reference rather than independent stereo ground truth. Per-backbone D7 margins come from one seed and the three-seed study bounds only the pooled effect.

b) What the metrics cannot say.: No-reference temporal consistency is gameable, which we show on D4D [21] — deformable surgical stereo whose geometric scale contract is unresolved, so we make no geometric claim on it and use it only as the substrate for this control. The module reduces the no-reference score by 53.9%, but copy-previous removes 97.3% and warped-previous scores exactly zero by construction. Pull-warp support detects out-of-frame sampling but not internal disocclusion.

c) What does not improve downstream.: SCARED-C supplies corrected per-frame poses, so geometry can be accumulated in one world frame and the disagreement between views of a point measured: what a map fuses rather than what a frame scores. On held-out D7 refinement reduces that disagreement in $4/4$ sequences and for both heads, by 11.7% of the excess over a ground-truth floor of 0.06–0.21 mm (9.9% for the 142-channel ablation). On development D2 it *increases* it by 15.5% (7.1% for the ablation). The splits differ in how much there is to remove — raw excess is 0.55–1.11 mm on D7 against 0.18–0.22 on D2 — which suggests alignment recovers geometry a map can fuse where the raw prediction is poor, and costs more than it gains where the raw already sits near the reference floor. Mapping benefit is conditional, not general. The reversal is not an artefact of one estimator. Repeating the measurement across five backbones, two of them excluded from temporal-head training, gives 23 cells in which the sign never changes: all 15 D2 cells are negative (-32.1 to -5.4%) and all 8 D7 cells positive ($+8.7$ to $+15.4\%$). What that does *not* settle is the mechanism. The two raw-excess ranges stay disjoint by a factor of two (0.171–0.290 mm against 0.552–1.113), because no backbone makes D2 as hard as D7, and within D2 the ordering does not hold either — the same estimator loses 32.1% at 0.269 mm of excess and 16.5% at 0.290. We therefore report the condition as robust across estimators and do not claim raw difficulty as its cause.

d) Cost, and where it is.: The head is compact but the deployed system also runs frozen stereo and two SEA-RAFT passes: 28.90 ms overhead at 95.2 MB peak VRAM on an H100. The frozen estimator dominates that budget by up to an order of magnitude, so the end-to-end rate is set by the backbone, from 11.6 FPS on Fast-FoundationStereo (overhead 33.5%) to 1.4 FPS on StereoAnywhere (4.0%); *no real-time claim is made*.

Stage timing locates it, on an otherwise idle H100: the two SEA-RAFT passes are 26.2 of the 28.9 ms, against 0.67 for the evidence tensor and 1.00 for the head. The evidence tensor is that cheap because TETHER builds no learned features; the 142-channel ablation spends 5.13 ms there, a 13.8% difference in the total. The reverse SEA-RAFT pass exists only to compute C_t^{FB}, which reaches the head as one of the six motion channels, so dropping that pass would free 13.1 of the 28.9 ms — 45% of the module — if the cue can be spared.

It can. Replacing it with a constant at inference, the trained head otherwise untouched, moves the D2 closure from 3.69% to 3.74% at 1.0 and 3.79% at 0.5, worsening 4 and 3 of the fifteen cells: not a cost but a small gain, and one an order of magnitude inside the ± 0.38 point seed spread, so the honest reading is that the cue contributes nothing this head uses. The 142-channel configuration behaves oppositely — the same substitution costs it 0.069 and 0.101 points of its own 4.046%, with 12 of 15 cells worse — which is consistent with the rest of the ablation: the learned evidence made the head sensitive to a cue that the geometric evidence space renders redundant. We report this rather than ship it, since it is measured on development only and on a head trained with the cue present; a head trained without it is the experiment we have not run. We evaluate perception geometry, not downstream control, and demonstrate neither clinical safety nor patient benefit.

VIII. Conclusion

TETHER attaches one causal temporal module, trained once, to frozen and heterogeneous stereo estimators without touching their internals or using future frames. Its gain is not temporal averaging: on development D2 it reduces EPE by 3.69% against 0.59% for the best matched fixed blend and 0.87% for the best tuned unlearned residual rule under an identical state machine — and those blends buy mean error with bad pixels, the best fixed one costing 2.64% more of them, a trade the learned gate does not make — while direct flow-confidence weighting and unbounded recurrence degrade accuracy, the latter by 19.17%. A frozen $H = 4$ checkpoint improves all five estimators on held-out D7 in EPE, Bad1 and RMSE at once, including two excluded from training, recovers 3.8–$11.1\times$ more Bad1 pixels than it introduces, and holds at $5.84 \pm 0.22\%$ Bad1 across three seeds. Zero-shot transfer to DRENDS improves all 25 recording–backbone cells on every metric.

Five results bound what may be claimed. The no-reference consistency metric is won outright by a policy that merely replays warped history. The fusion weight inverts its relation to harm between splits, so it is an intervention magnitude, not an uncertainty. Development gains can carry selection optimism that held-out gains do not, and the evidence space decides whether they do: the 142-channel configuration spans 5.03 points of D2 effect across seeds against 0.69 on D7, while the shipped head spans 0.60 and 0.40. A module that declares a support must apply it to what it emits, a contract almost invisible in domain that costs the whole metric-depth tail out of it. And world-frame reconstruction consistency improves on the held-out split while degrading on development, for every backbone we tried, so mapping benefit is conditional on something we can demonstrate but not yet name.

What remains open is where to intervene: the oracle over the module's own action space is far from reached, and the head's weight is a magnitude, not a ranking.

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